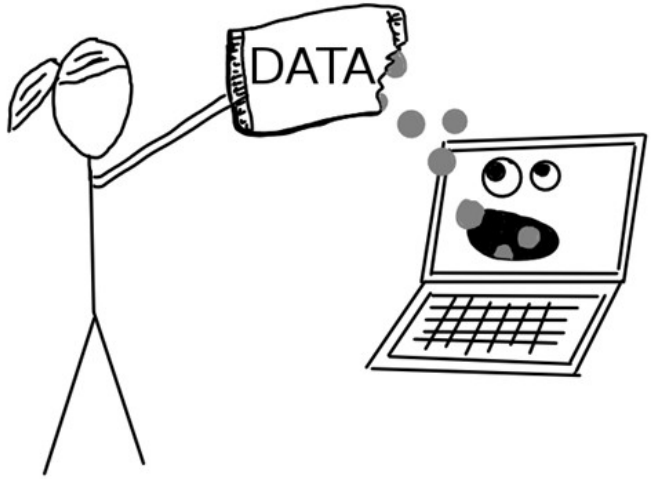

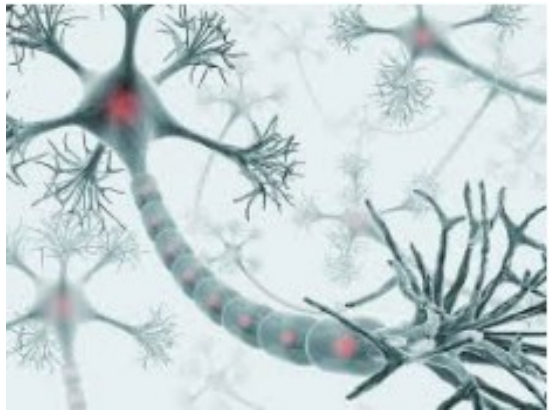

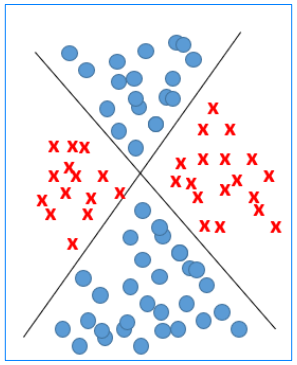

Multilayer Neural Network
non linearly separable problem

2023/2024

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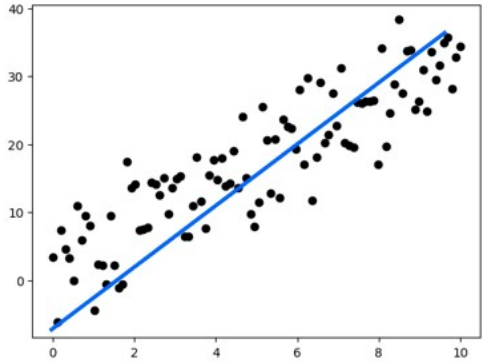
Abdelwahab.naji@gmail.com
ENSAO, Université Mohamed premier - Oujda
https://www.youtube.com/c/AbdelwahabNaji

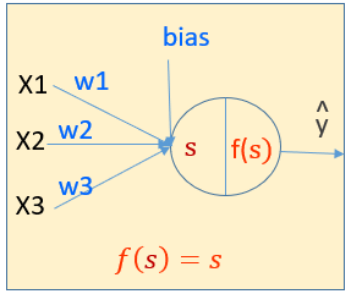
Neural Network for Supervised Learning

1

Simple Linear Regression Problem



Random point cloud

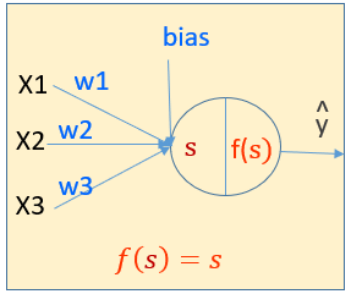


2

Multi Variate Linear Regression Problem

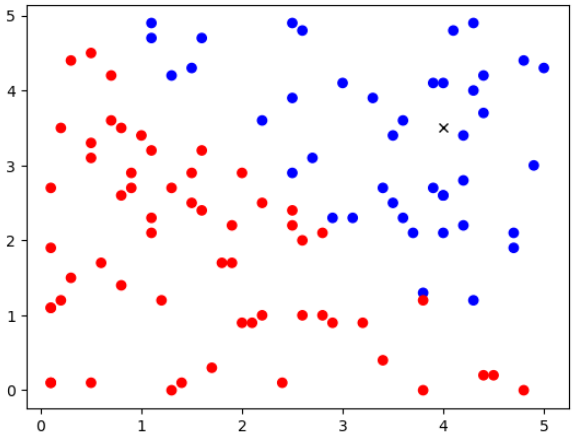
TV	Radio	Newspaper	CA
230.1	37.8	69.2	**22,1**
44.5	39.3	45.1	**10,4**
17.2	45.9	69.3	**9,3**
...			
199.8	2.6	21.2	**10,6**

Advertising p

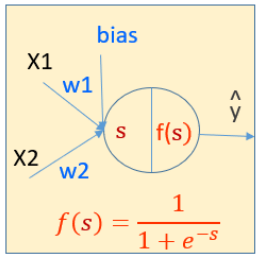


3

Linearly Separable problem

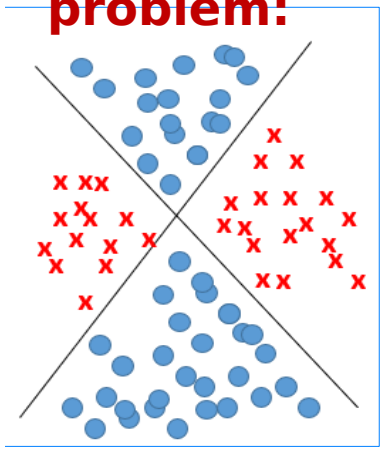


Red or Blue?

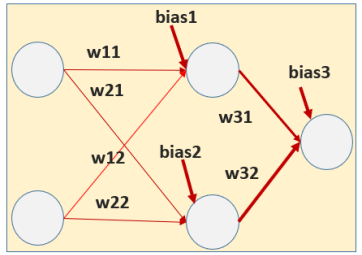


4

Non Linearly Separable problem:

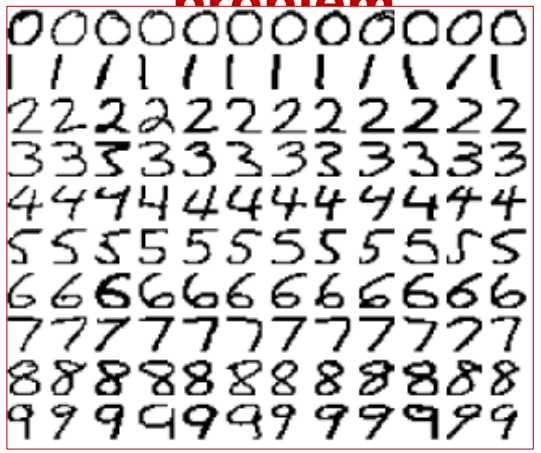


XOR Problem

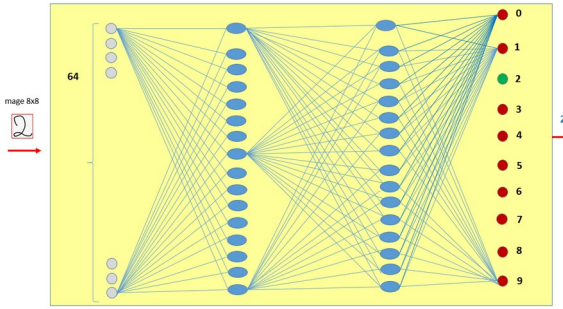


5

Complex classification problem

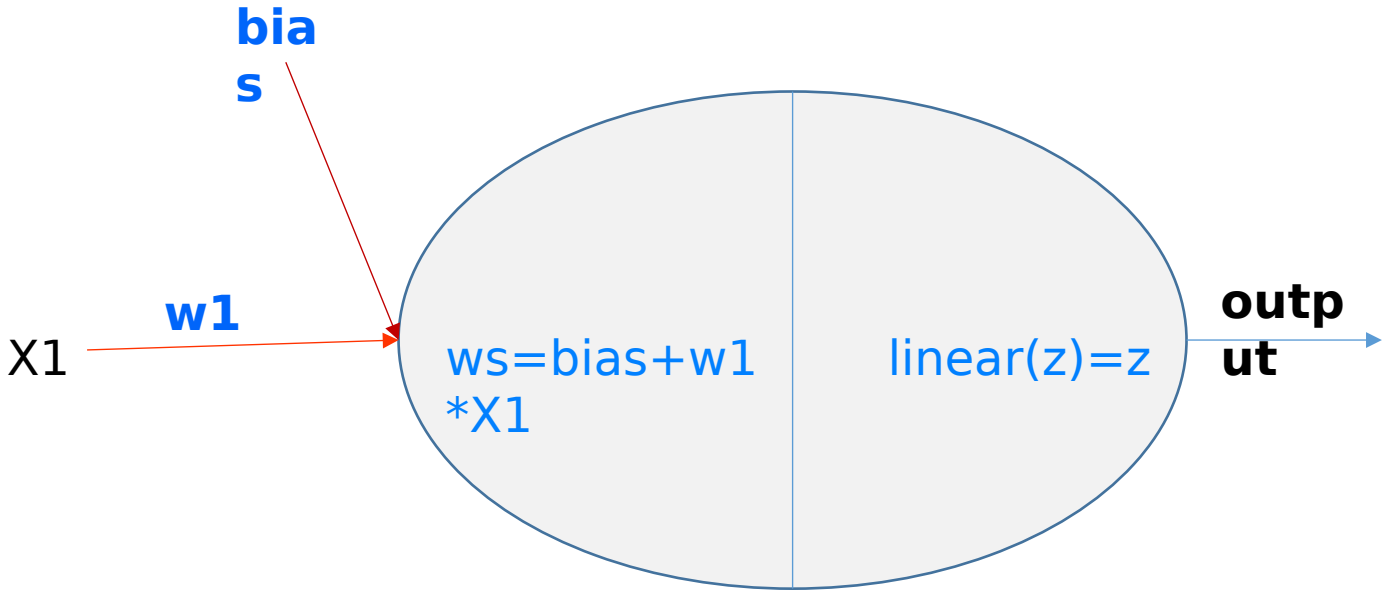


Hand Written digits



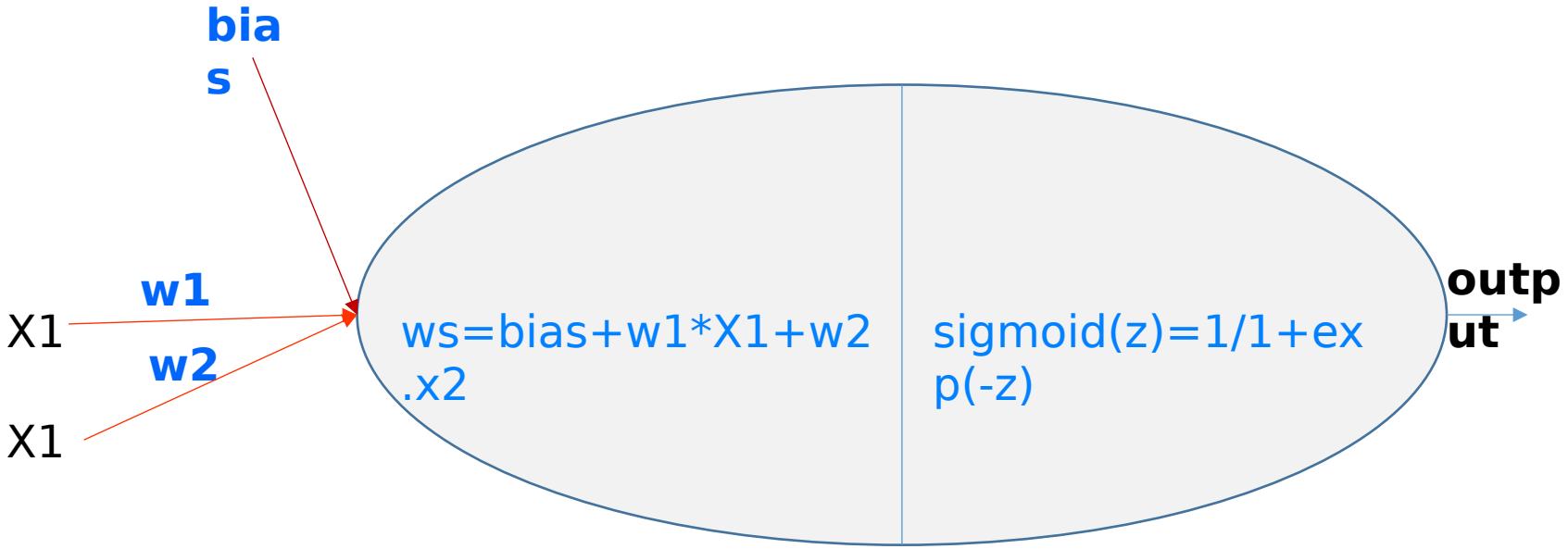
Neural Network – One Neurone

Simple Regression Problem



Neural Network – One Neurone

Simple Classification Problem



1

Simple Regression problem

```
#Librairies
import matplotlib.pyplot as plt
from random import seed
from random import
```

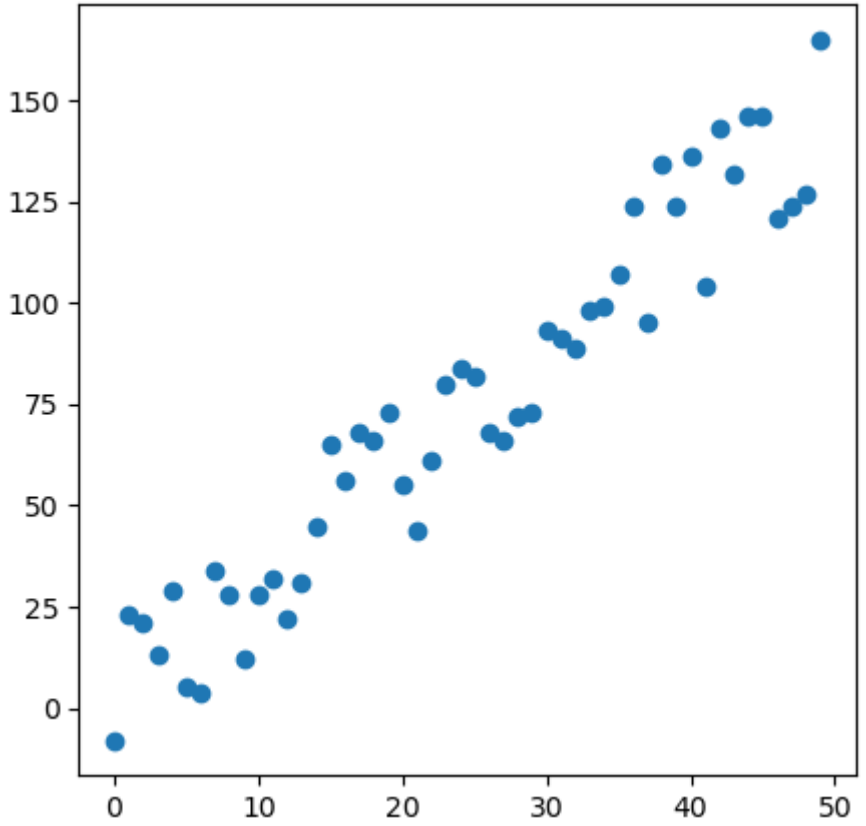
```
np.random.seed(0)
```

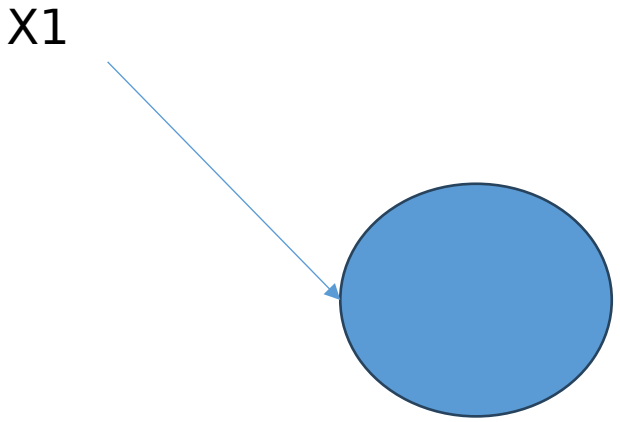
```
def f(X):
    return 1+3*X
```

```
def generateData (n):
    X=np.array([i for i in range(n)])
    y_model=f(X)
    y =np.array([i+randrange(-20,20) for i in y_model])
    return X,y_model,y
```

```
n_samples = 50
X,y_model,y=generateData(n_samples)
```

```
plt.figure(figsize=(5, 5))
plt.scatter(X, y)
plt.show()
```




X1

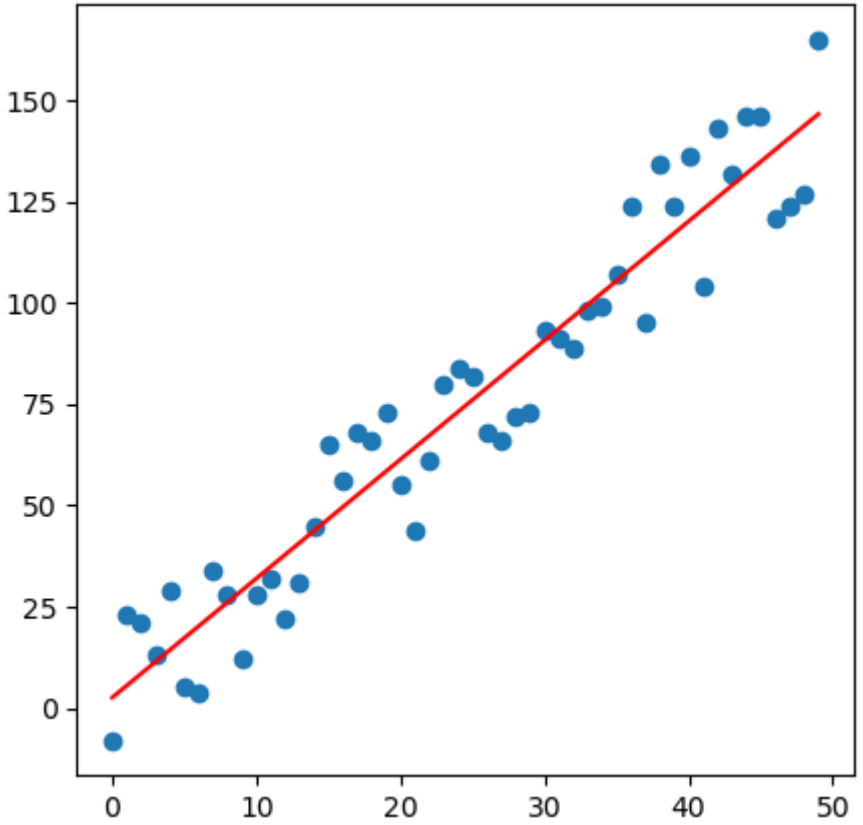
```
def fit(X,y):
    lr=0.001
    epochs=10000
    bias=0.5
    w1=0.5
    n=len(y)
    for i in range(epochs):
        py=bias+w1*X
        dbias=-(2/n)*sum(y-py)
        print(epochs,"---------------->",w1," , ",bias)
        dw1=-(2/n)*sum(X*(y-py))
        w1=w1-dw1*lr
        bias=bias-dbias*lr
    return bias,w1
```

```
bias,w1=fit(X, y)
```

```
def predict(X):
    return bias+w1*X
```

```
predicted_y=predict(X)
```

```
plt.figure(figsize=(5, 5))
plt.title("Simple Regression Model)", size=16)
plt.scatter(X, y)
plt.plot(X, predicted_y, c="red")
plt.show()
```



2

Simple classification problem Linearly Separable

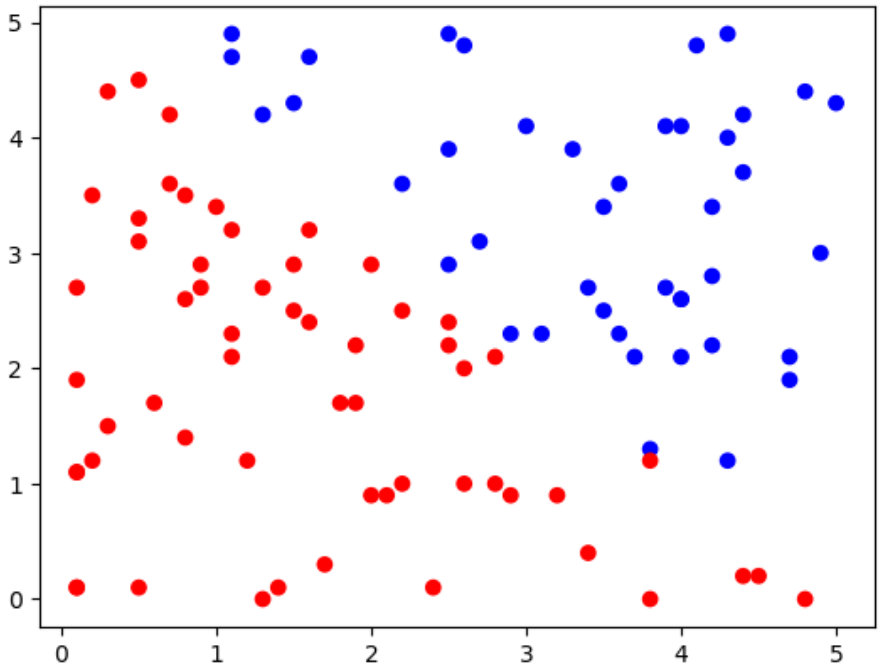
```
import matplotlib.pyplot as plt
import numpy as np
import matplotlib.colors as clr
```

```
def f(x):
    return -x+5
```

```
# generate random floating point values
from random import seed
from random import random
# seed random number generator
def generateData(n):
    seed(1)
    min=0
    max=5
    # generate random numbers between
0-1
    L=[]
    for _ in range(n):
        valueX = random()
        scaledvalueX = min + (valueX *
(max - min))
        valueY = random()
        scaledvalueY = min + (valueY *
(max - min))
        x=round(scaledvalueX,1)
        y=round(scaledvalueY,1)
        y0=f(x)
        cl=0
        if(y>y0):
            cl=1
```

```
L=generateData(10
0)
L=np.array(L)
X=L[:,0]
y=L[:,1]
```

```
colors = ['red','blue']
plt.scatter(L[:,0],L[:,1],c=L[:,2],cmap=clr.ListedColor
map(colors))
```



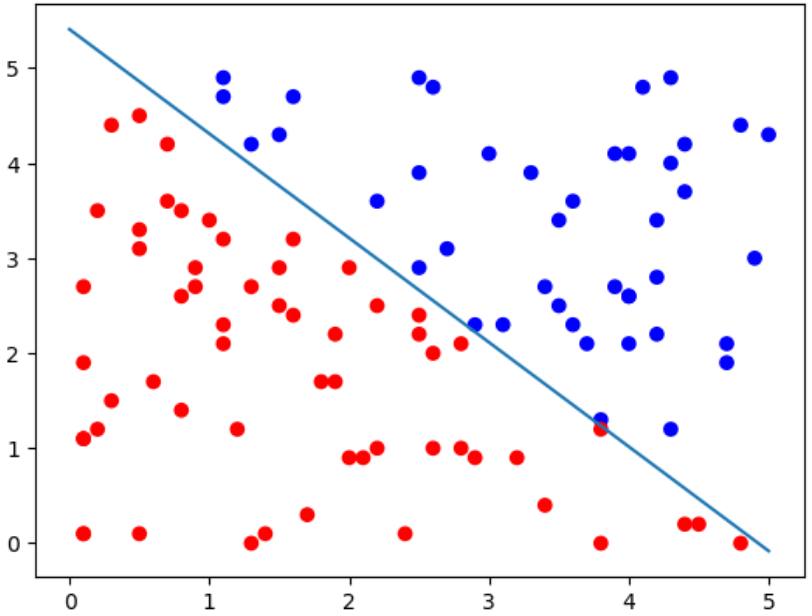
```
def fit(X,y):
    #initialisation
    n=len(y)
    bias=0.5
    w1=0.5
    w2=0.5
    epochs=5000#hyperparamter
    lr=0.01 #hyperparameter
    #learning
    for i in range(epochs):
        y=L[:,2]
        ws=bias+w1*L[:,0]+w2*L[:,1]
        py=sigmoid(ws)
        #update
        dbias=-(2/n)*sum(y-py)
        dw1=-(2/n)*sum(L[:,0]*(y-py))
        dw2=-(2/n)*sum(L[:,1]*(y-py))

        bias=bias-dbias*lr
        w1=w1-dw1*lr
        w2=w2-dw2*lr
    return bias,w1,w2
```

```
bias,w1,w2=fit(X,y)
```

```
def getY(x,bias,w1,w2):
    return -w1/w2*x-bias/w2
```

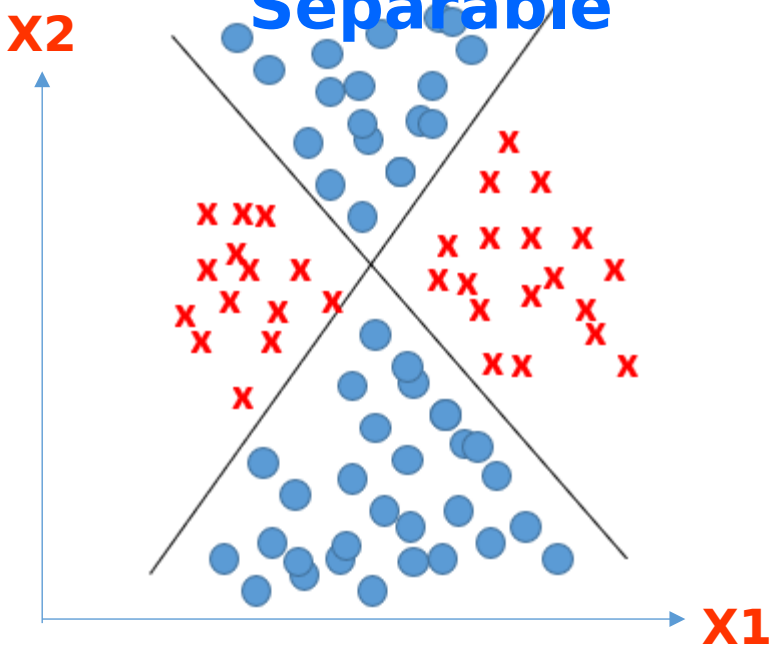
```
x1=0
x2=5
y1=getY(x1,bias,w1,w2)
y2=getY(x2,bias,w1,w2)
```

```
colors = ['red','blue']
plt.scatter(L[:,0],L[:,1],c=L[:,2],cmap=clr.ListedCol
ormap(colors))
#plt.plot([x1,x2],[y1,y2])
plt.plot([x1,x2],[y1,y2])
```



3

Complex classification problem non Linearly Separable



4

Images classification

Mise en œuvre de Neural Network

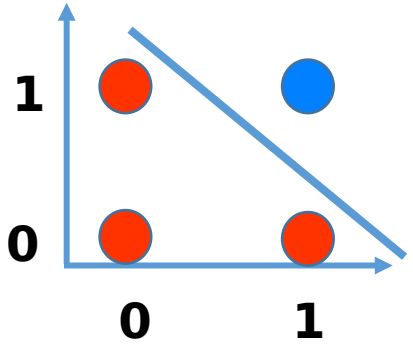
- **NN: Multi – Layer**
  - Présentation du problème **non linéairement séparable**
  - Implémentation du Problème **non linéairement séparable (**avec tensorflow, keras **)**
    - Exemple 1: simple XOR problem
      - Générer les données
      - Créer et entrainer un multilayer neuralnetwork
    - Exemple 2: general XOR problem
      - Générer les données
      - Créer et entrainer un multilayer neuralnetwork
    - Exemple 3: fashion classification problem
    - Exemple 4: hand written digits classification problem
  - Implémentation from scratch
    - Exemple: fashion classification problem

Problem

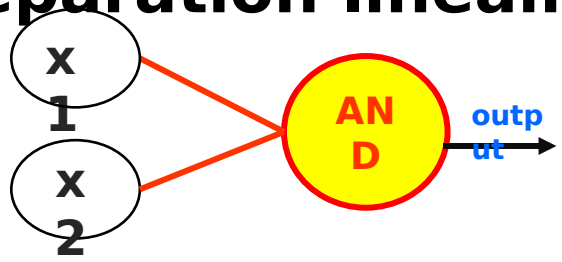
Séparation de
AND, OR,NAND, NOR, XOR

And

	F	T
T	F	T
F	F	F

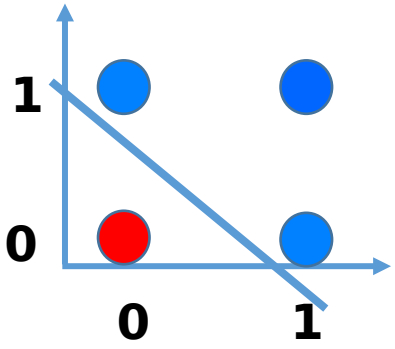


On peut avoir une séparation linéaire

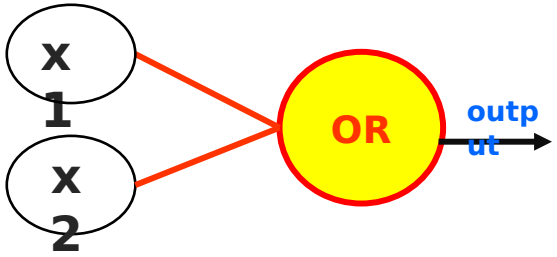


OR

	F	T
T	T	T
F	F	T

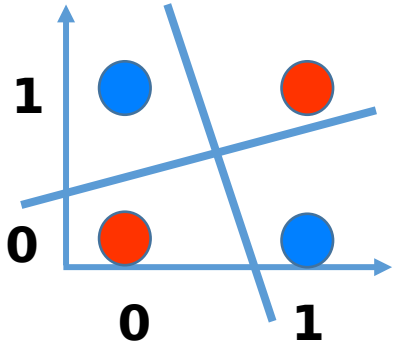


On peut avoir une séparation linéaire



XOR

	F	T
T	T	F
F	F	T

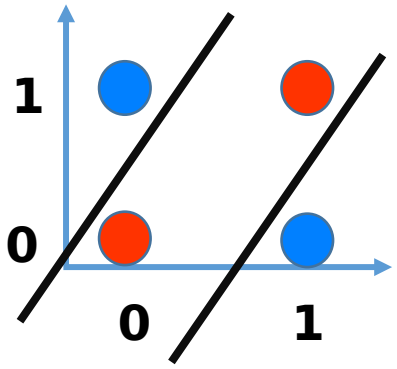


On ne peut pas avoir une séparation linéaire

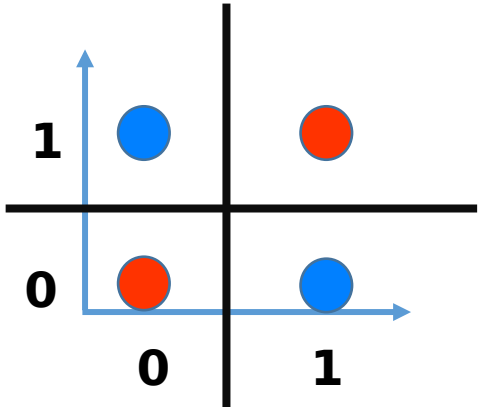
Séparation de X0R

XOR

	F	T
T	T	F
F	F	T

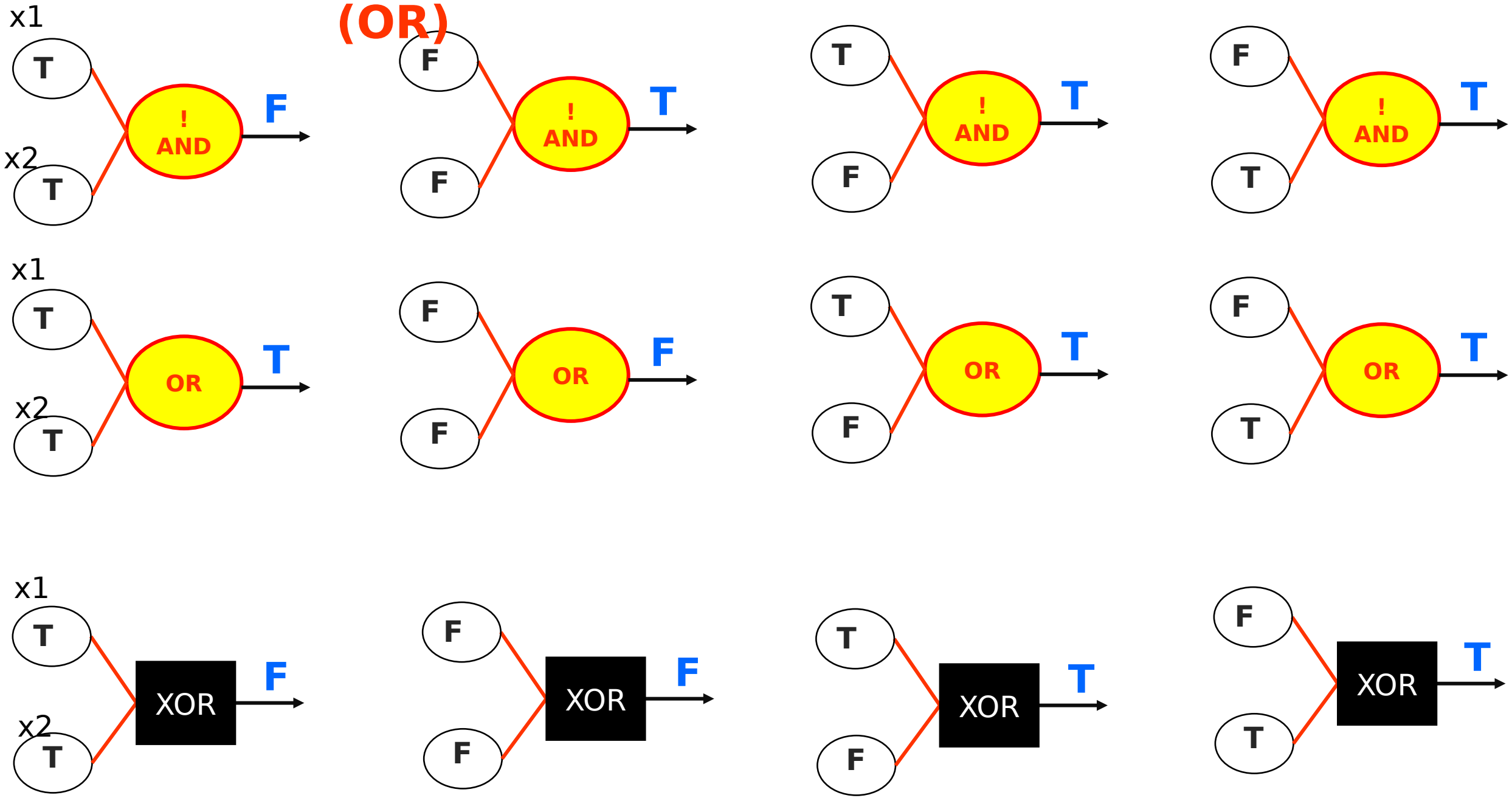


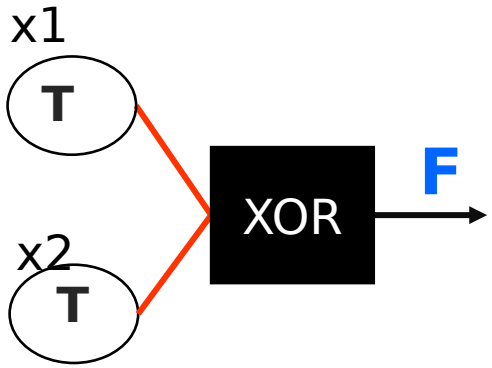
On peut utiliser plus qu'une ligne pour faire la séparation

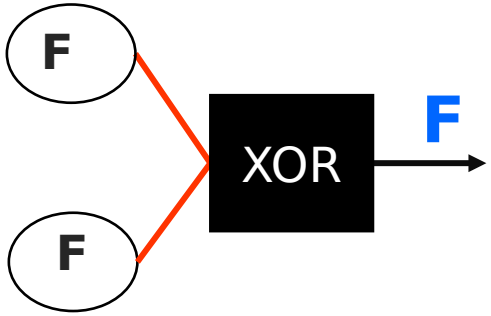


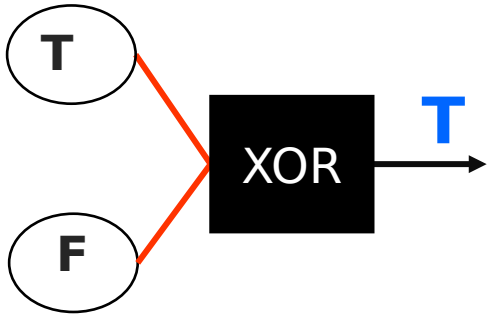
XOR = (!AND) and (OR)

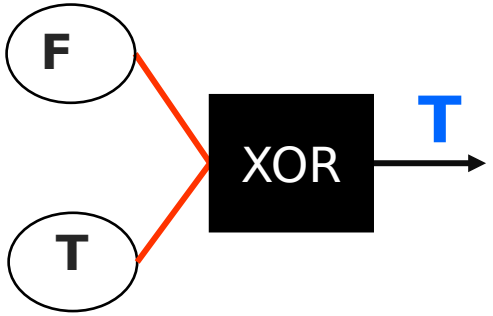
XOR = (!AND) and (OR)

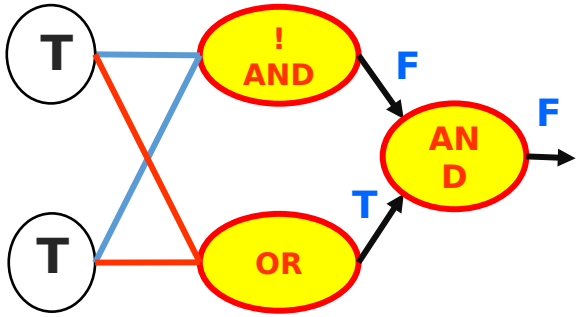


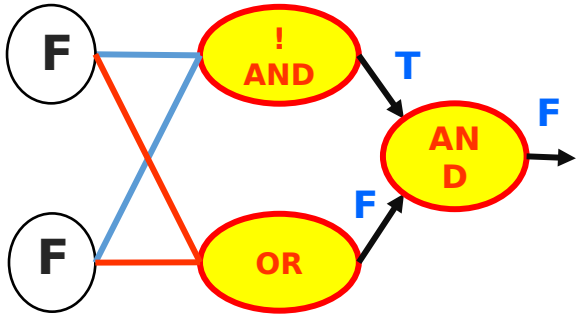

x1
T
x2
T
XOR
F

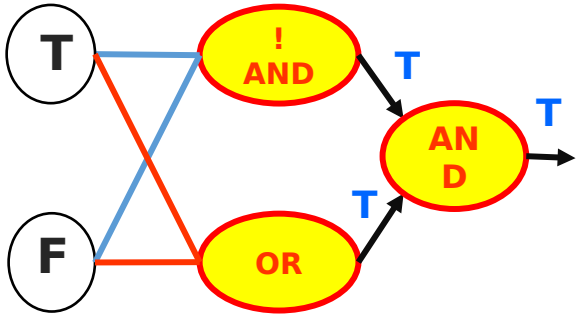

F
F
XOR
F

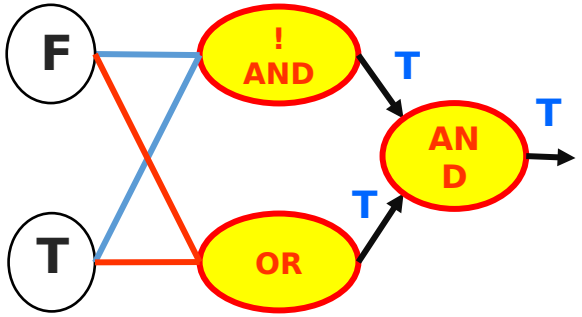

T
F
XOR
T

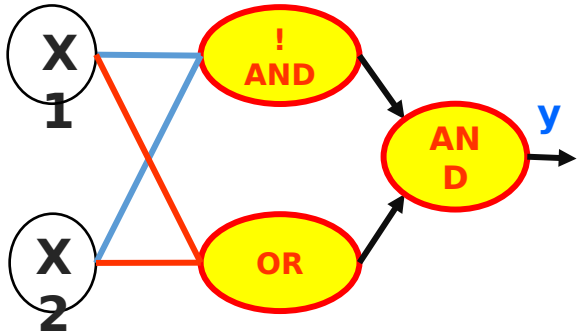

F
T
XOR
T


T
T
! AND
OR
F
T
AN D
F


F
F
! AND
OR
T
F
AN D
F


T
F
! AND
OR
T
T
AN D
T


F
T
! AND
OR
T
T
AN D
T


X 1
X 2
! AND
OR
AN D
y

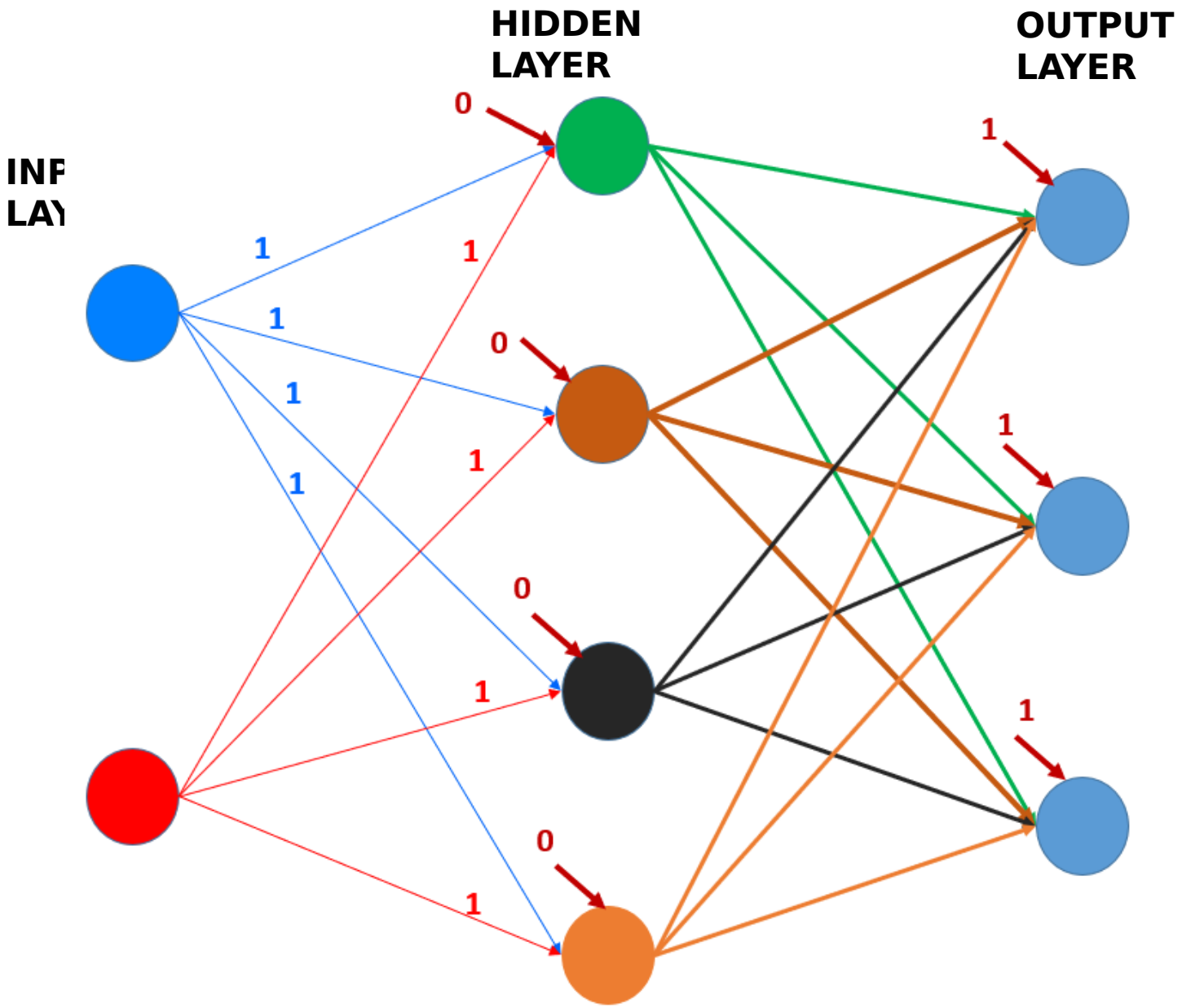
Utiliser keras pour créer un réseau de neurone avec plusieurs neurones (couche, layer)

Ce qu’il faut connaitre:

1. **Comment créer un multilayer neural network (en général)?**
2. Découvrir les éléments du neural network créé: coeficients, bias

Neural Network Model et ses paramètres

1. Architecture
2. Créer le modèle
3. Créer une couche et l’ajouter au modèle
4. Paramètres d’une couche


HIDDEN LAYER
OUTPUT LAYER
INP
LAY
0
1
1
1
1
1
1
0
1
1
0
1
1
0
1

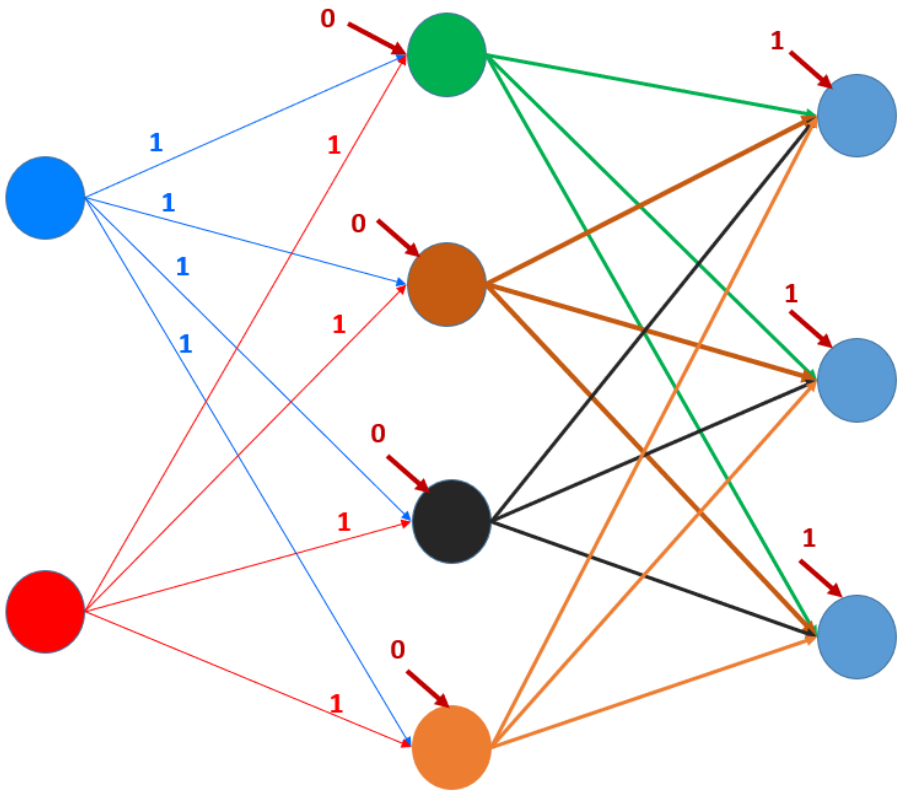
Model et ses paramètres

```
from keras.models import Sequential
from keras.layers import Flatten, Dense, Activation
```

```
model = Sequential()
```

```
#input_layer=Flatten(input_shape=[2, 1])
#model.add(input_layer)
hidden_layer=Dense(4,
input_dim=2,kernel_initializer='ones',bias_initializer='zeros
')
model.add(hidden_layer)
output_layer=Dense(3,
kernel_initializer='random_uniform',bias_initializer='ones')
model.add(output_layer)
print(model.summary())
```

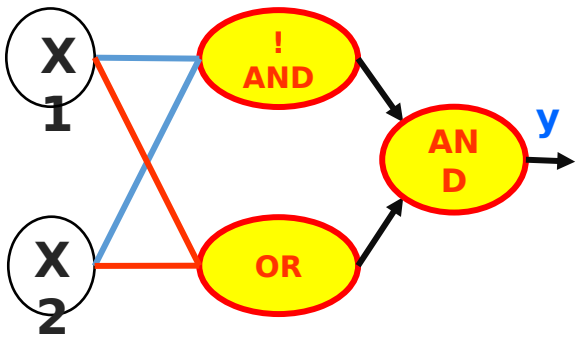
```
model.compile(loss="categorial_crossentropy",
            optimizer="adam",
            metrics=["accuracy"])
```



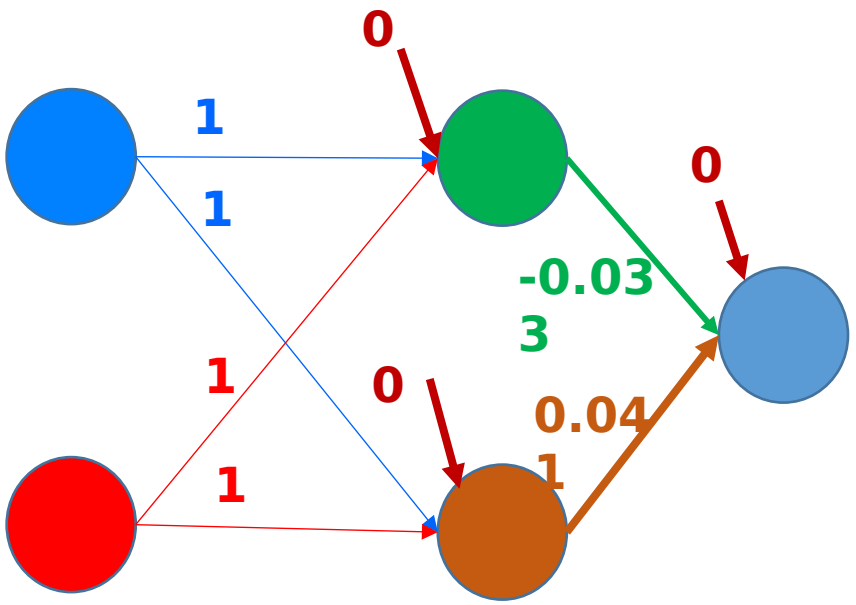

```
[array([[1., 1., 1., 1.],[1., 1., 1., 1.]],
dtype=float32),
       array([0., 0., 0., 0.], dtype=float32),
 array([[-0.03575971, -0.02119735,
0.01558626],
     [ 0.04222678, -0.04578313,  0.0360521 ],
     [-0.02753322,  0.00745941,
-0.04211374],
     [-0.04299896, -0.04190966,
0.02975568]], dtype=float32),
     array([1., 1., 1.], dtype=float32)]
```

Revenons à XOR Problem

Les étapes à suivre pour créer le modèle



1. **Data**
2. **Model**
   - Create the model, compile the model: keras, tensorflow
     - Optimizer function: adam
     - Loss function: binary_crossentropy
   - Train the model
   - Parameters
3. **Prediction**
4. **Evaluation**



1. Dataset

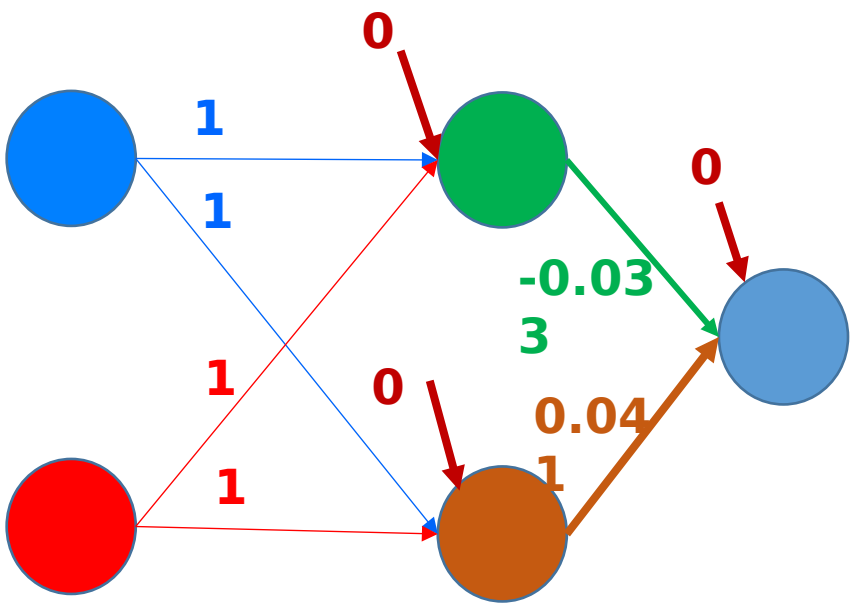
```
X=np.array([[0,0],[0,1],[1,0],[1,1]])
y = np.array(np.logical_xor(X[:, 0] > 0, X[:, 1] >
0), dtype=int)
```

2. Model

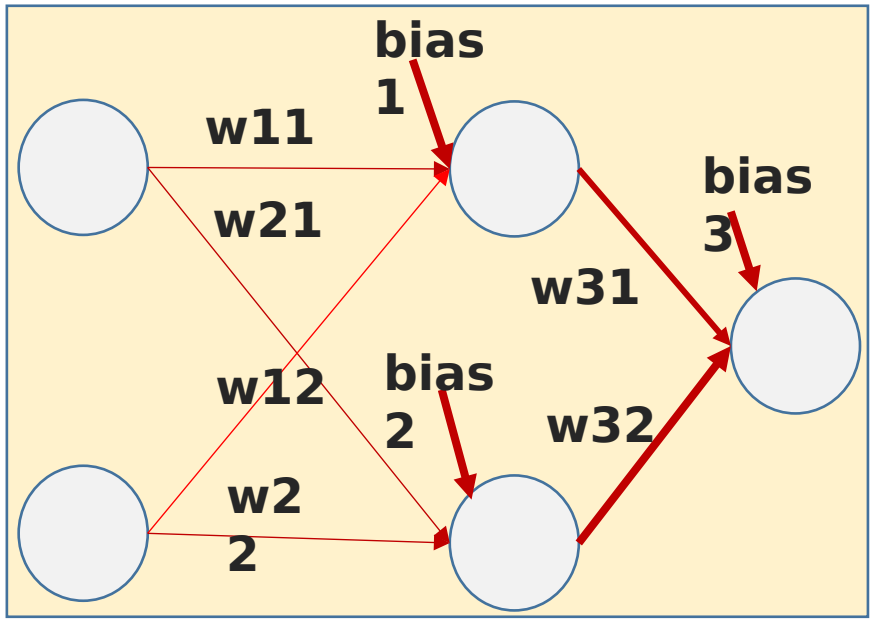
```
from keras.models import Sequential

from keras.layers import Activation, Dense,Flatten

model = Sequential()

#input_layer = Flatten(input_shape=[2,1])

hidden_layer = Dense(2,input_dim=2,
kernel_initializer='ones',bias_initializer='zeros', activation='relu')

output_layer =
Dense(1,kernel_initializer='random_uniform',bias_initializer='zero
s', activation='sigmoid')

#model.add(input_layer)

model.add(hidden_layer)

model.add(output_layer)

coef=model.get_weights()

print(coef)
```

```
model.compile(loss='binary_crossentropy',optimizer='adam',metri
cs=['accuracy'])
```

```
model.fit(X,y,epochs=1000)

print(model.get_weights())

print(model.predict(X)
```



[array([[1., 1.], [1., 1.]], dtype=float32), array([0., 0.], dtype=float32), array([[-0.03330588], [0.04190588]], dtype=float32), array([0.], dtype=float32)]


bias
1
w11
w21
bias
3
w31
w12
bias
2
w32
w2
2

```
model.fit(X,y,epochs=1000)
print(model.get_weights())
print(model.predict(X)
```

2. Model (avec une autre fonction d'activation)

```
from keras.layers import LeakyReLU
act = LeakyReLU(alpha = 0.3)
```

```
model = Sequential()
hidden_layer = Dense(2,input_dim=2, kernel_initializer='ones',bias_initializer='zeros',activation=act)
model.add(hidden_layer)
output_layer = Dense(1,kernel_initializer='random_uniform',bias_initializer='zeros',activation='sigmoid')
model.add(output_layer)
model.add(hidden_layer)
model.add(output_layer)
coef=model.get_weights()
print(coef)
```

```
model.compile(loss='binary_crossentropy',optimizer='adam',metri
cs=['accuracy'])
```

```
model.fit(X,y,epochs=1000,batch_size
=4)

print(model.get_weights())

print(model.predict(X))
```